Public Capital Investment Frictions: A Piece in the Euler Equation Puzzle

Yichu Li*

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Abstract

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*The London School of Economics and Political Science. y.li395@lse.ac.uk.

1 Introduction

Much of development economics research attempt to understand what is holding back economic development in lower- and middle-income countries. The categorisation of countries into 'developing' versus 'developed' implies that the former have higher development potentials. The Solow (1956) growth model predicts exactly this pattern because poorer economies with smaller capital stocks should face higher marginal returns to capital investment and therefore grow faster than richer economies. While the broad pattern of higher GDP growth rates in lower-income countries over the past four decades is consistent with this prediction, many countries have still believed to been growing at rates below their potentials.

Lucas (1990) and Caselli and Feyrer (2007) points out that the higher marginal product of capital in lower-income countries is often balanced-off by their lower endowment in complimentary factors like human capital, or lower total factor productivity. This provides an aggregate-level explanation for why capitals do not flow from higher-income to lower-income countries as much as expected. However, the puzzle is not fully addressed because on the aggregate-level, it does not explain why lower-income countries do not invest in human capital or R&D to benefit from the high capital returns. Moreover, through micro-level RCTs, de Mel et al. (2008) show that microenterprises in Sri Lanka face 55%-63% real anual returns to capital, and Balboni et al. (2022) show significant long-run returns on agricultural capitals sufficient to push households in Bangladesh out of a poverty trap. Kremer et al. (2019) uses an household-level intertemporal constrained optimisation model to illustrate an '*Euler Equation Puzzle*' where people in lower-income settings should experience a 20% consumption growth rate facing a 150% gross return, but the actual consumption growth rates are much lower.

Relating to this paradox, a large body of literature including Balboni et al. (2022) and research covered in Banerjee (2003) point to liquidity and credit constraints facing households and firms; some including Acemoglu et al. (2001), Mauro (1995) and Banerjee et al. (2002) point to institutional ineffectiveness and insufficient property rights protections; while Kremer et al. (2019) point to studies in behavioural channels such as present bias. However, the existing literature focus primarily on private investment channels. For example, these are scenarios where farmers investing in a cattle or fertilisers and benefit fully from the output gains. Meanwhile, returns on certain capitals like infrastructure or human capital are also high in lower-income countries. Investment in public capitals often require fiscal spending from the government funded primarily by tax revenues. Besley and Persson (2013) show that in lower-income countries taxation is more difficult administratively while frictions against efficient spending are stronger. The constraints on both private and public capital investment seem to be likely correlated with the existing capital accumulation and development levels, hinting towards an endogenous capital investment mechanism that relate to the poverty trap and convergence literature. Studies like Balboni et al. (2022) shows that poverty traps likely bind the ultra-poor households to invest, while research on aggregate growth like Quah (1996) documents the bimodal pattern and Aghion et al. (2005) identifies the role of financial

development in threshold convergence.

Nonetheless, Kremer et al. (2019) questions why households would not rationally attempt more to smooth consumption or leverage on informal credits when motivated by the high returns. The poverty traps thresholds are likely less binding for the non-ultra-poor, but they systematically not exploit the capital returns sufficiently either. The same question extends to the public capital investment: it seems irrational for above-subsistence households in lower-income countries to avoid taxes even with low expected penalties because the long-term returns from public capitals can be sufficiently high. This paper adapts and develop on the baseline mode of Kremer et al. (2019) to show that the set-up of the taxation-to-government-spending mechanism produces a collective action problem at the household level. Public capital investment requires highly lumpy fiscal spending funded by taxation, and the total returns are shared by each household like public goods. When the perceived state taxation capacity is low, households expect low actual returns on public capital investment via taxation because they cannot credibly expect others to pay taxes symmetrically. Therefore, for sufficiently low state taxation capacity, households' dominant strategy is to consume all and pay zero taxes.

2 A Household Intertemporal Constrained Optimisation Model on Capital Investment

I build on the intertemporal optimisation framework of Kremer et al. (2019). I first recall the model setup and present the Euler Equation Puzzle through the model. I then consider the extreme scenario where only public capitals exist and investment occurs only through taxation, showing that this generates a collective action problem and a government efficiency problem that dilutes the effective return to capital when state taxation capacity and efficiency is low. Finally, I will extend this analysis into a generalised model incorporating both private and public capital, allowing households to allocate investments between the two.

2.1 Baseline: Standard Euler Equation

Consider in a deterministic discrete-time intertemporal consumption model, households maximise their discounted lifetime utility over an infinite time horizon.

Optimisation Problem

Households face the following optimisation problem:

$$\max_{\{C_t\}} \sum_{t=0}^{\infty} \delta^t u(C_t), \tag{1}$$

where $0 < \delta < 1$ is the annual discount factor, C_t is the household consumption in period t, and $u(C_t)$ measures the utility from consuming C_t in period t. $u(C)$ is increasing and concave and

follows Inada Conditions: $u'(C) > 0$, $u''(C) < 0$, $\lim_{C\to 0} u'(C) = \infty$, and $\lim_{C\to\infty} u'(C) = 0$.

Budget Constraint

While the model set-up in Kremer et al. (2019) consider the gross return to capital, I consider the return to capital which allows me to model capital depreciation that the PWT data provides at the aggregate level. Therefore, the budget constraint is:

$$x_{t+1} = [(1-d) + f'(K_t)](x_t - C_t), \tag{2}$$

x_t is household assets ('cash on hand') in period t with x_0 given, C_t is consumption in period t, and $f'(K_t)$ is the return to capital, and d is the capital depreciation rate. $f(K)$ is increasing and sconcave, following: $f'(K) > 0$, $f''(K) < 0$, $\lim_{K\to 0} f'(K) = \infty$, and $\lim_{K\to\infty} f'(K) = 0$ (long-run steady state).

Discrete Time Euler Equation

Solve for optimal intertemporal consumption path:

$$u'(C_t) = \delta \cdot [(1-d) + f'(K_t)] \cdot u'(C_{t+1}), \tag{3}$$

and assume constant-elasticity-of-substitution (CES) utility:

$$u(C) = \begin{cases} \frac{C^{1-\sigma}}{1-\sigma} & \text{if } \sigma \neq 1 \\ \ln(C) & \text{if } \sigma = 1 \end{cases}$$

so that the elasticity of intertemporal substitution is constant and independent of C_t. This gives the consumption growth rate:

$$\frac{C_{t+1}}{C_t} - 1 = (\delta[(1-d) + f'(K_t)])^{\frac{1}{\sigma}} - 1, \tag{4}$$

where $\frac{1}{\sigma}$ is the elasticity of intertemporal substitution.

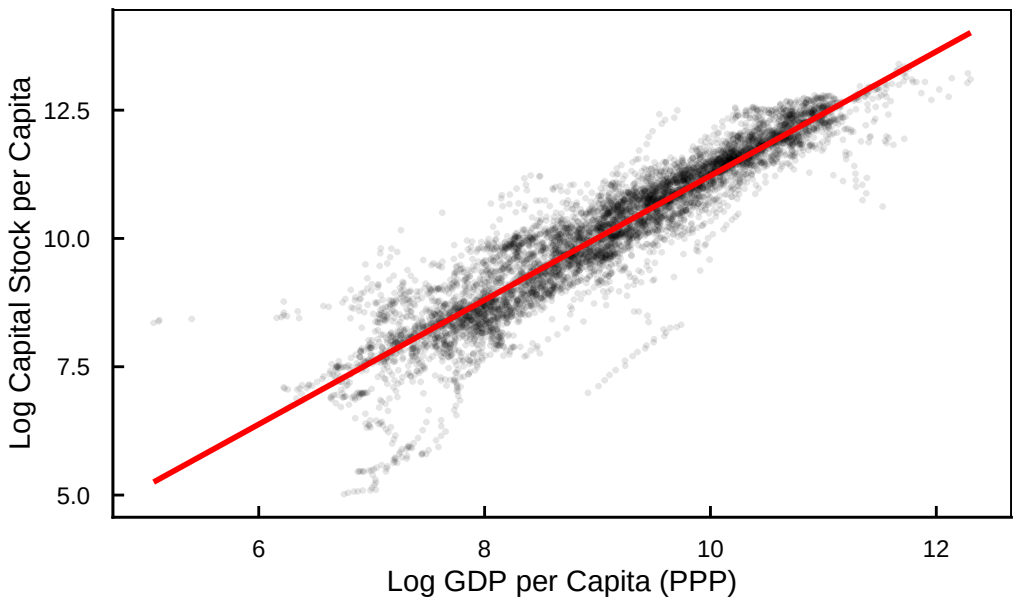
The Puzzle

Assume that the model governs the intertemporal consumption behaviour of a representative household, the model should also apply at the macro level. To illustrate the puzzle with macro-level data, I use $\delta = 0.96$ [1]and $\sigma = 2$ [2] proposed in Kremer et al. (2019), and internal rates of return (IRR) from PWT [3] (Feenstra et al., 2015) as a proxy for $f'(K)$[4] to model the predicted consumption

[1] $\frac{1}{\delta} - 1 \approx 0.04$, implying a 4% annual rate of time preference.
[2] Elasticity of intertemporal substitution of 0.5.
[3] Available for download at www.ggdc.net/pwt
[4] See https://www.rug.nl/ggdc/docs/the_next_generation_of_the_penn_world_table.pdf

growth rates under Equation 4. Intuitively, a concave production function means that countries with lower capital stock face higher returns on capital investment. Figure 1 shows that countries with lower income have lower capital stock which is expected because the Solow (1956) model shows that output increases with capital accumulation. Therefore, according to Equation 4, consumption growth rates should be high in lower-income countries. However, Figure 2 Panel (A) shows that in many countries, the actual growth rates is insensitive to IRR and many fall below the model predictions. Moreover, Figure 2 Panel (B) shows a perplex trend where countries with higher predicted consumption growth rates tend to have larger gap between predicted and actual rates. This pattern is most pronounced in countries lower-middle and low income countries. Therefore, the core of the puzzle are why households in these economies invest at levels far below what the standard model predicts as optimal, and why is this pattern more pronounced in countries where returns on capitals are high.

Figure 1: Lower Capital Stock in Lower-Income Countries

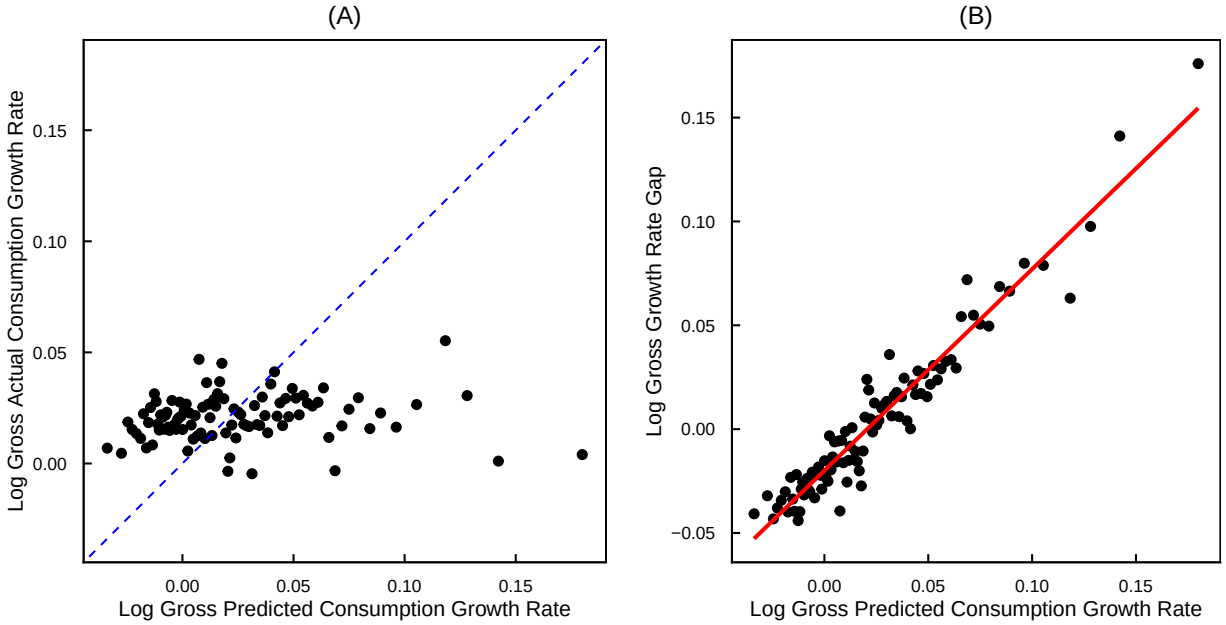


Notes: Red line is OLS fit. Panel data from 1950 to 2023. Capital stock in millions US$ at 2021 PPPs. Output−side real GDP in millions US$ at 2021 chained PPPs. Source: PWT 11.0.

2.2 Endogenous Capital Accumulation

There is a potential concern for the findings in Figure 2 that the insensitivity of the actual consumption growth rate towards the model prediction signals severe model mis-specification. This would mean that the Euler equation model setup in Section 2.1 cannot capture the true underlying mechanism of consumption growth from capital investment, at least under aggregate data. However, the orthogonal polynomial regressions in Table 1 show significant quadratic correlation between the actual and predicted log gross consumption growth rate, while the linear term correlation is not statistically significant . This suggests that a non-trivial relationship between predicted and actual growth does exist. Nonetheless, the small R^2 across specifications shows that the baseline

Figure 2: Actual vs. Predicted Consumption Growth



Notes: Binned scatter plots with 100 equal−frequency bins. The blue dashed line is the 45 degree line. The red line is OLS fit. Countries with populations below 1 million and irregular internal rates of return excluded. Data after 2019 is excluded to avoid Covid−19 distortions. Source: PWT 11.0.

model explains only a tiny fraction of the variation in actual consumption growth.[5] Therefore, understanding the puzzle will require identifying the factors that systematically prevent households from exploiting high capital investment returns according to the baseline model prediction.

A mathematically plausible way to understand the reason for Figure 2 Panel (B) is to consider that households or countries that tend to under-invest in capital likely simultaneously have lower consumption growth and lower capital stocks. This means that the capital return rates should be high due to low capital stocks which predict high consumption growth, but the actual consumption growth is low because they invest sub-optimally, therefore the consumption growth gap is large. The countries with higher predicted consumption growth in Figure 2 Panel (B) tend to be countries that under-invest relative to the predicted levels, have lower capital stocks, and exhibit larger consumption growth gaps. Table 2 also shows that the linear pattern is predominantly driven by large linear term coefficient while the quadratic coefficient is statistically significant but too small to be visible in the binned scatter plot.

While this intuition still does not resolve the puzzle in Figure 2 Panel (A) of why some countries systematically under-invest given high returns, it provides an important insight for the likely endogeneity in the capital accumulation process that is consistent with the quadratic correlation in Table 1. Existing literature has studied extensively sources of frictions such as credit market or institutional inefficiencies that hinder optimal capital investment given its returns. The intuition above suggests that these frictions are likely endogenous to the capital stock level, where

[5]Appendix Table A1 reports significantly higher R^2 after adding per captia GDP growth and TFP growth as controls, suggesting more variations are explained by these factors.

Table 1: Polynomial Fit of Actual on Predicted Consumption Growth

	Log Actual Consumption Growth		
	Linear	Quadratic	Cubic
Log Predicted Growth	0.027		
	(0.025)		
Poly term 1		0.083	0.083
		(0.076)	(0.076)
Poly term 2		−0.259***	−0.259***
		(0.076)	(0.076)
Poly term 3			−0.084
			(0.076)
Intercept	0.020***	0.021***	0.021***
	(0.001)	(0.001)	(0.001)
Sample Size	6413	6413	6413
R^2	0.0002	0.0020	0.0022

* $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$

Notes: Orthogonal polynomial regressions of log actual on log predicted gross consumption growth. Standard errors in parentheses.

Table 2: Polynomial Fit of Gap on Predicted Consumption Growth

	Log Actual Consumption Growth		
	Linear	Quadratic	Cubic
Log Predicted Growth	0.973***		
	(0.025)		
Poly term 1		3.005***	3.005***
		(0.076)	(0.076)
Poly term 2		0.259***	0.259***
		(0.076)	(0.076)
Poly term 3			0.084
			(0.076)
Intercept	−0.020***	0.004***	0.004***
	(0.001)	(0.001)	(0.001)
Sample Size	6413	6413	6413
R^2	0.1940	0.1955	0.1956

* $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$

Notes: Orthogonal polynomial regressions of log gap (actual - predicted) on log predicted consumption growth. Standard errors in parentheses.

more frictions exist in low-capital countries and the frictions cause them to accumulate capital slowly and persist with low capital stocks. Moreover, the quadratic correlation form in Table 1 indicates that the countries on the lower and higher ends of predicted consumption growth rate experience lower actual growth, while the countries in the medium range experience the higher growth.

2.3 Capital as a Public Good

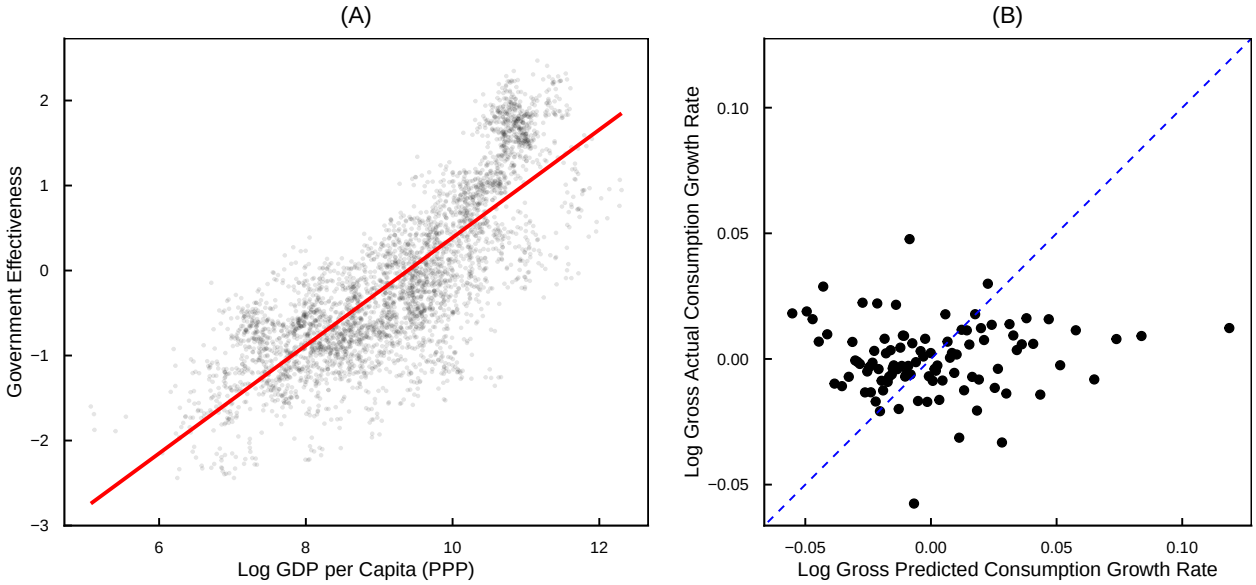
The baseline model in Section 2.1 assumes capital investments are private for households. This means that households receive the full returns $f'(K)$ because they would own the invested capital in productions. However, capitals with public good features like infrastructure often instead require public investment commonly through government spending, and the benefits from these investments are shared among the population. Aschauer (1989) estimates a high average elasticity of 0.39 between output and public investment in the United States over 1949-1985, while Bom and Ligthart (2014) proposes a more modest estimate of 0.106. The latter estimate that the short-run national-level elasticity of 0.083 still imply a 16% marginal return to public capital in OECD countries. The returns are likely higher in regional levels and much higher in lower-income countries with lower public capital stocks. However, Besley and Persson (2013) show likely political economy reasons including political instability, favourtism, inequality, or corruptions which undermine the taxation-to-government-spending mechanism necessarily for optimal public capital investment. These factors are more prominent in lower-income countries, while countries might improve taxation capacities and have reduced spending frictions over development.

However, if public capital investments yield high returns, why would rational households even not pay taxes voluntarily but choose to avoid them when possible? Indeed, findings in Mauro (1995) also show low public (and private) investment levels when governments are inefficient, and Figure 3 Panel (A) shows that government effectiveness is lower in lower-income countries. Therefore, this high government spending friction could discourage households to pay taxes if they believe the revenue is not put in use efficiently. Nonetheless, Figure 3 Panel (B) shows that the trend in Figure 2 Panel (A) persist even controlling for government effectiveness. My following model will present that the household level, the fiscal features of lower-income countries can also create a collective action problem so that they optimally avoid paying taxes individually, thereby producing a global sub-optimal outcome.

To illustrate, I will present an extreme case where we assume capitals are all publicly invested through taxation by governments. Consider the same optimisation problem as Equation 1 with a different budget constraint for homogeneous households in an economy. For any household i, the budget constraint is:

$$x_{i,t+1} = \frac{1}{N} \cdot [(1-d) + f'(K_t)] \cdot [(x_{i,t} - C_{i,t}) + \sum_{\forall j \neq i} (x_{j,t} - C_{j,t})], \tag{5}$$

Figure 3: Gap Persists after Controlling for Government Effectiveness



Notes: Red line is OLS fit. Panel data from 1996 to 2023. Output−side real GDP in millions US$ at 2021 chained PPPs. Panel B shows binned scatter plots with 100 equal−frequency bins after partialling out government effectiveness. The blue dashed line is the 45 degree line. Source: PWT 11.0, WGI. Source: WDI.

where N is the total number of households in the economy, and $\sum_{\forall j \neq i}(x_{j,t} - C_{j,t})$ is the aggregate public investment from all other households. Because public capitals can only be invested collectively, the total returns depend on the aggregate investment level. For simplicity, assume each individual household receive an equal share of the total returns, public capital investment is the only government expenditure, and the government has perfect efficiency.

Weak Taxation Capacity

First consider a scenario where the state taxation capacity is weak. To reflect weak taxation capacity, assume that household i makes public investment decision through taxation voluntarily and independent of other households' decisions. From the perspective of household i, the total returns from public capital investment as a public good is diluted among the population. Alternatively, for a fixed level of total returns on public capital, the impact of the household's investment level $(x_{i,t} - C_{i,t})$ on its assets in the next period is small for large N.

Solving for optimal intertemporal consumption path for household i given $\sum_{\forall j \neq i}(x_{jt} - C_{jt})$:

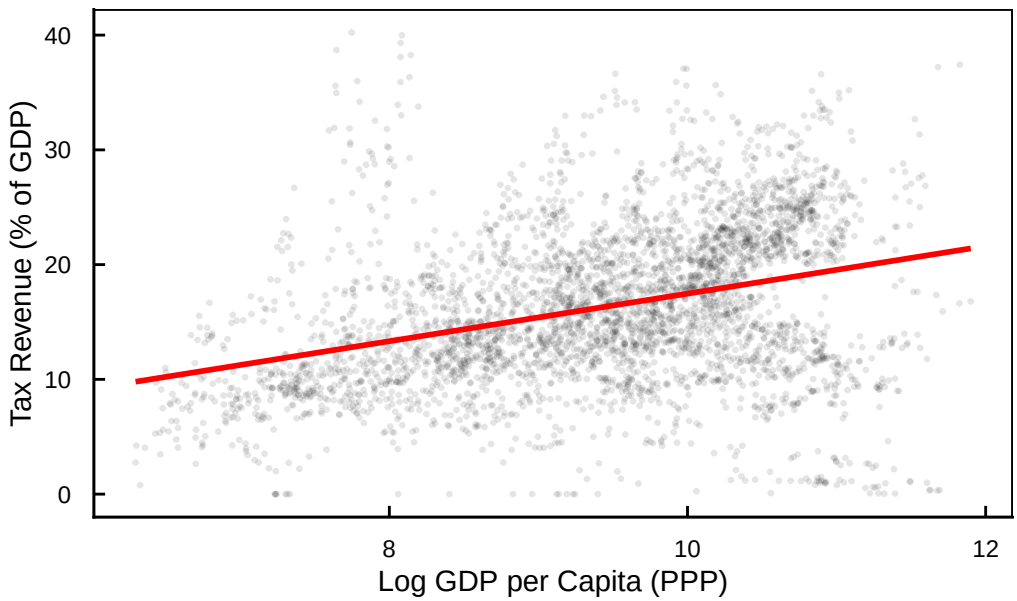
$$u'(C_{i,t}) = \frac{1}{N} \cdot \delta \cdot [(1-d) + f'(K_t)] \cdot u'(C_{i,t+1}). \tag{6}$$

Notice that when $N = 1$, this equation is identical to Equation 3, but $u'(C_{i.t})$ decreases as N becomes larger. For increasing and concave utility function, $u'(C_{i,t})$ is smaller when $C_{i,t}$ is larger. This means that household i will consume more and invest less as N increases. For sufficiently large N so that the Euler Equation cannot be satisfied by an interior solution, household i will optimally choose a corner solution of consuming all and invest none. $\frac{1}{N}$ produces a collective

action problem where the global optimal decision is for all households to invest and benefit from high $f'(K)$ but individuals households have a dominant strategy of not investing.

Figure 4 shows the countries with lower GDP per capita have lower tax revenue relative to their GDP. Therefore, this scenario is more consistent with lower-income settings and it is consistent with the reluctant tax compliance attitude among many low- and middle-income settings. This framework also provides an explanation to the prima facie irrationality of tax avoidance in lower-income countries where people can benefit largely from public investments.

Figure 4: Higher Tax Revene in Higher-Income Countries



Notes: Red line is OLS fit. Panel data from 1972 to 2023. Output−side real GDP in millions US$ at 2021 chained PPPs. Source: WDI.

Strong Taxation Capacity

Higher income countries tend to have stronger taxation capacity because, for example, income tax can be more easily collected from employees (Jensen, 2022). Although, public investment is less voluntary for tax rates as given, households might make public investment decisions by influencing the tax rates through political channels. The key difference is that they are more likely to expect other households to also comply with the decided upon tax rates.

Consider a simple case of homogeneous households facing the same optimisation problem as Equation 1 but assume instead perfect enforcability upon any chosen taxation level and perfect government efficiency. Any individual household i chooses a taxation level (public investment level) $(x_{i,t} - C_{it})$ and expect symmetry from other households. Because households are homogeneous, the following should hold:

$$[(x_{i,t} - C_{i,t}) + \sum_{\forall j \neq i} (x_{j,t} - C_{j,t})] = N \cdot (x_{i,t} - C_{i,t}),$$

so that the budget constraint becomes identical to Equation 2:

$$x_{i,t+1} = \frac{1}{N} \cdot [(1-d) + f'(K_t)] \cdot N \cdot (x_{i,t} - C_{i,t}) = [(1-d) + f'(K_t)](x_{i,t} - C_{i,t}).$$

Solving for optimal consumption path will produce an Euler Equation identical to Equation 3. Notice that the collective action problem is resolved in this setting because the perfect tax enforcement enables households to optimise consumption conditional on the expectation of symmetrical behaviours from others.

2.4 A Generalised Static Model

The collective action problem explored above is an example channel to explain why public capital investment frictions exist. In reality, there are likely many interacted channels and the specific functional form is difficult to define. Even for the collective action model, it is very possible that the severity of the problem is not linearly sensitive to the population size in a country. Nonetheless, the public capital investment frictions at the household-decision level can be generalised as a wedge $\phi_{pub} \in [0, 1]$ where lower values indicate larger frictions. Moreover, recall that the literature also highlights various channels for the private-side frictions, so that the private-side frictions can be also generalised by $\phi_{priv} \in [0, 1]$ analogously. A generalised intertemporal household opitmisation model with both private and public capital returns has the following budget constraint:

$$x_{t+1} = [(1-d) + r_t^{eff}] \cdot (x_t - C_t), \tag{7}$$

where

$$r_t^{eff} = (1-\tau) \cdot \phi^{priv} \cdot f'(K_t^{priv}) + \tau \cdot \phi^{pub} \cdot f'(K_t^{pub}),$$

with K^{priv} denoting private capital stock and τ denoting the share of the total investment amount that the household chooses to invest in public capital. Notice that households choose τ freely to maximise their budget constraint. Because $f'(K_t^{priv})$ and $f'(K_t^{pub})$ are concave, this means that at equilibrium:

$$\phi^{priv} \cdot f'(K_t^{priv}) = \phi^{pub} \cdot f'(K_t^{pub})$$

and rearranging:

$$\frac{\phi^{pub}}{\phi^{priv}} = \frac{f'(K_t^{priv})}{f'(K_t^{pub})} \tag{8}$$

shows that the ratio between the public and private capital investment frictions can be given by the ratio between the marginal investment returns from private and public capital.

Solving for optimal consumption path under the above constraint produces an Euler Equation:

$$u'(C_t) = \delta \cdot [(1-d) + r_t^{eff}] \cdot u'(C_{t+1}),$$

and the consumption growth rate is given by:

$$\frac{C_{t+1}}{C_t} - 1 = \{\delta[(1-d) + r_t^{eff}])\}^{\frac{1}{\sigma}} - 1. \tag{9}$$

Recall the pattern illustrated in Figure 2 Panel (B): countries with higher predicted consumption growth rates tend to exhibit larger gaps between predicted and actual growth. Under the above model, the interpretation is that the combined frictions from both channels are higher in lower-income countries with high returns on both types of capital so that households rationally under-invest because the expected capital returns are lower than the realised returns.

3 Accounting the Public Capital Investment Frictions

The theoretical static model features four structural parameters:

$$\Theta = \{\delta, \sigma, \tau, \phi\},$$

where δ is the annual discount factor, σ is the risk aversion coefficient and $\frac{1}{\sigma}$ is the elasticity of intertemporal substitution, τ is the share of total investment in public capital, and ϕ is the severity of the collective action problem for public capital investment. To explore the data under the static model, I discipline the model by exogenously fix $\delta = 0.96$ and $\sigma = 2$ proposed by Kremer et al. (2019) based on the literature, same with the exercise in Figure 2. Although in the theoretical model, τ is chosen by the households, the public capital investment shares from ICSD can be interpretated as the equilibrium outcome of this choice.

To use macroeconomic data, I proxy for the returns on private and public capitals households face by the following Cobb-Douglas production function facing a representative household in the economy:

$$Y_{it} = A_{it}(K_{it}^{priv})^{\alpha^{priv}}(K_{it}^{pub})^{\alpha^{pub}}L_{it}^{1-\alpha^{priv}-\alpha^{pub}},$$

where the first-order conditions are:

$$f'(K_{it}^{priv}) = \alpha_{priv} \cdot \frac{Y_{it}}{K_{it}^{priv}} \quad \text{and} \quad f'(K_{it}^{pub}) = \alpha_{pub} \cdot \frac{Y_{it}}{K_{it}^{pub}}. \tag{10}$$

I use the calibrate using the results from Bom and Ligthart (2014) to fix the output elasticity of public capital while allowing the output elasticity of private capital to move mechanically. For the short-run output elasticity of private capital, assuming for simplicity the labour market is competitive in aggregate:

$$w_{it} = \frac{\partial Y_{it}}{\partial L_{it}} = (1 - \alpha_{short}^{pub} - \alpha_{it}^{priv})\frac{Y_{it}}{L_{it}},$$

so that:

$$\alpha_{short,it}^{priv} = 1 - \alpha_{short}^{pub} - \left(\frac{wL}{Y}\right)_{it}, \tag{11}$$

where the PWT provides the data on the labour share of income which is equivalent to the output

elasticity of labour. Using the conservative national short-run estimate of 0.083 for α^{pub}, this approach derives a pooled mean output elasticity of labour of 0.523, and a pooled mean ouput elasticity of private capital of 0.394. Bom and Ligthart (2014) merge the cointegrated relationship variable and models featuring long differences variables and estimate the mean long-run elasticity of 0.122 based on this composite variable. This is consistent with the nature of longer time horizons for public projects like infrastructures and therefore longer time horizon for the returns to materialise. To capture this nature, the long-run elasticity of private capital should be calculated over every 5 year non-overlapping horizon:

$$\alpha^{priv}_{long,i,[t,t+4]} = 1 - \alpha^{pub}_{long} - \left(\frac{wL}{Y}\right)_{i,[t,t+4]}, \tag{12}$$

and this derives a pooled mean output elasticity of labour of 0.518, and a pooled mean output elasticity of 0.36.

3.1 Difficulty to Isolate the Public Channel

Ideally, ϕ^{pub} may be approximated by a variable that is associated with the public capital investment channel but not with the private channel, and Section 2.3 and Besley and Persson (2013) show that it should likely relate to government taxation capacities. The self-employment share in total employment is a natural candidate in this context. Jensen (2022) identifies that the income tax exemption threshold co-moves with the employee/self-employed composition over development, such that the tax base consistently covers segments where third-party enforceable employee income predominates (85–95% employee share). This pattern is consistent with Table 3 Panel (A) which shows higher self-employment percentage in total employment significantly associates with lower log public capital investment across specifications. However, Table 3 Panel (B) shows a highly similar pattern for the private capital investment channel. This means either self-employment share is correlated with both the private and public capital investment returns through channels like financial institution, or the frictions from both sides are highly correlated. Notice that under Equation 8, correlation between the private and public capital returns implies correlation between the frictions from both sides. In either case, self-employment share is likely not a good proxy to distinguish ϕ^{pub} from the overall investment friction. Identifying isolated effects from specific channels is commonly challenging using aggregate data. Therefore attempting alternative identification strategies is beyond the scope of this paper. Nonetheless, this empirical finding motivates a follow-up test on the correlation between both friction channels.

3.2 Limited Cross-Channel Frictions Spillovers

Countries with high self-employment shares tend to be lower-income countries that share characteristics like prevalent liquidity constraints, property rights insecurity, or institutional ineffectiveness that could affect both the private and public capital investment frictions simultaneously. If this were the case, it would be a mistake to treate the two channels separately in any empirical work.

Table 3: Self-Employment and Capital Investment

	Pooled	Country FE	Year FE	Country + Year FE
Panel A: $\log I_{it}^{pub}$				
Self-Employment (%)	-0.044***	-0.069***	-0.043***	-0.027*
	(0.002)	(0.014)	(0.002)	(0.013)
Sample Size	4674	4674	4674	4674
Within R^2		0.129	0.615	0.018
Panel B: $\log I_{it}^{priv}$				
Self-Employment (%)	-0.051***	-0.073***	-0.051***	-0.026*
	(0.002)	(0.014)	(0.003)	(0.011)
Sample Size	4674	4674	4674	4674
Within R^2		0.159	0.640	0.021

Notes: Dependent variable is log per capita investment. Standard errors clustered at the country level in parentheses. $^{*}p < 0.05$, $^{**}p < 0.01$, $^{***}p < 0.001$.

Computing the marginal capital returns according to Equation 10 and test for dynamic relationship between them is informative of the correlation between the frictions from each side under Equation 8. This is because the capital returns change due to investment behaviours correlated to investment frictions from the two channels, therefore dynamic relationship between private and public capital returns can signal correlation between the friction channels. However, one caveat is that dynamic relations can also occur due to the upward bias from common trends that do not necessarily create associations between the friction channels, but this also means that the real association is likely below the estimated levels.

Table 4 shows Granger tests on first-differenced [6] log short-run marginal returns from private and public capital. Changes in public capital marginal returns exhibit significant positive autocorrelation (lagged own coefficient 0.17–0.23), while the private side does not. This asymmetry suggests slower adjustment in public investment, meaning that public capital investment frictions are sticker, while the market adjusts faster on the private side. This is consistent with factors like political inertia, long infrastructure project duration, discontinuous government budgets which do not impact private-side investments. There are statistically significant small spillovers from the public capital investment frictions to the private side, but the reverse are even smaller in magnitude and less robust across specifications. The tiny within R^2 under country and year fixed effects also indicate that lagged cross-channel and own-channel returns explain a small fraction of within-country variation in marginal returns at annual frequency.

Table 5 shows the Granger tests results for long-run private and public capital returns over five-year windows. The coefficients on own-channel returns are larger and more significant compared

[6] Im-Pesaran-Shin panel tests fail to reject the null of a unit root for both $\log f'(K^{priv})$ ($p = 0.28$) and $\log f'(K^{pub})$ ($p \approx 1.00$) in levels, but strongly reject in first differences ($p < 0.001$ for both).

Table 4: Short-Run Granger Tests

	Pooled	Country FE	Year FE	Country + Year FE
Panel A: $\Delta log f'(K_{it}^{priv})$				
$\Delta log f'(K_{i,t-1}^{pub})$	0.093***	0.104***	0.086***	0.097***
	(0.023)	(0.024)	(0.023)	(0.024)
$\Delta log f'(K_{i,t-1}^{priv})$	0.014	-0.015	0.006	-0.023
	(0.018)	(0.018)	(0.018)	(0.018)
Sample Size	6417	6417	6417	6417
Within R^2		0.005	0.005	0.004
Panel B: $\Delta log f'(K_{it}^{pub})$				
$\Delta log f'(K_{i,t-1}^{pub})$	0.228***	0.186***	0.214***	0.167***
	(0.018)	(0.018)	(0.018)	(0.018)
$\Delta log f'(K_{i,t-1}^{priv})$	-0.037**	-0.022	-0.044**	-0.027*
	(0.014)	(0.014)	(0.014)	(0.014)
Sample Size	6417	6417	6417	6417
Within R^2		0.028	0.032	0.020

Notes: Short-run specification uses Bom-Ligthart (2014) short-run elasticity 0.083. First difference removes AR(1) structure. Standard errors clustered at the country level in parentheses. $^{*}p < 0.05$, $^{**}p < 0.01$, $^{***}p < 0.001$.

to the short run across all specifications. The stickier own-channel persistence in the long run is likely due to the five-year window reflecting more long-term structural economic trends like industrial, institutional, or TFP trajectories. However, although the private-to-public spillovers are still small, the magnitude almost doubles and the shrinking significance levels are likely resulted from shrinking sample size. This is likely also related to the coefficient absorbing the long-term trends. The long-run public-to-private spillovers attenuate instead. The increasing within R^2 under country and year fixed effect indicate that the long-run model has a better explanatory power.

Recall that the cross-channel spillovers can be over-estimated based on the current models by omitting common investment and structural economic trends. Nonetheless, the small magnitude of cross-channel spillovers likely indicate that the correlation between the two investment channels are likely small. Under the country and year fixed effects, both short-and long-run spillover coefficients are not significant when controlling for total factor productivity, while the Granger tests on log first-differenced capital accumulation level estimate the largest coefficient of public-to-private spillovers of 0.177 ($p < 0.1$) which is still small in magnitude.[7] The overall empirical evidence suggest although spillovers between the private and public friction channels exist, the two investment channels and by extension the frictions operate largely via separate mechanisms. Therefore, studying the public and private capital investment channels separately is important based on this finding.

[7]See Appendix Table A2.

Table 5: Long-Run Granger Tests

	Pooled	Country FE	Year FE	Country + Year FE
Panel A: $\Delta log f'(K_{it}^{priv})$				
$\Delta log f'(K_{i,[t-5,t-1]}^{pub})$	-0.062	0.061	-0.045	0.083
	(0.045)	(0.052)	(0.045)	(0.052)
$\Delta log f'(K_{i,[t-5,t-1]}^{priv})$	0.365***	0.147**	0.361***	0.133**
	(0.041)	(0.049)	(0.041)	(0.048)
Sample Size	1079	1079	1079	1079
Within R^2		0.037	0.108	0.039
Panel B: $\Delta log f'(K_{it}^{pub})$				
$\Delta log f'(K_{i,[t-5,t-1]}^{pub})$	0.461***	0.341***	0.472***	0.350***
	(0.041)	(0.048)	(0.041)	(0.048)
$\Delta log f'(K_{i,[t-5,t-1]}^{priv})$	-0.080*	-0.048	-0.083*	-0.057
	(0.037)	(0.045)	(0.037)	(0.044)
Sample Size	1079	1079	1079	1079
Within R^2		0.089	0.157	0.090

Notes: Long-run specification uses 5-year averaged variables and Bom-Ligthart (2014) national long-run elasticity 0.122. Year FE is applied according to the 5-year windows. First difference removes AR(1) structure. Standard errors clustered at the country level in parentheses. $^{*}p < 0.05$, $^{**}p < 0.01$, $^{***}p < 0.001$.

3.3 Ratios between Private and Public Capital Returns

It is difficult to decompose consumption growth induced by private and public capital investment. However, recall the equilibrium condition of Equation 8, I can rewrite:

$$\frac{\phi_{it}^{pub}}{\phi_{it}^{priv}} = \frac{f'(K_{it}^{priv})}{f'(K_{it}^{pub})} = \frac{\alpha^{priv}}{\alpha_{it}^{pub}} \left(\frac{K_{it}^{pub}}{K_{it}^{priv}} \right)$$

Using the above condition and the national and short-run output elasticity of public capital 0.083 in Bom and Ligthart (2014), Figure 5 shows most countries across time have larger public-side wedge compared to the private side, where the marginal returns on private capital investment is higher than public capital investment. This is an indication that the private capital investment frictions are higher on average across levels of development. However, the red OLS fit shows that this pattern is more prominent on the lower log per capital GDP end, and it attenuates towards the higher end. This pattern is also robust after controlling for country and year fixed effects.

Using instead the five-year specification and long-run output elasticity of public capital of 0.122,

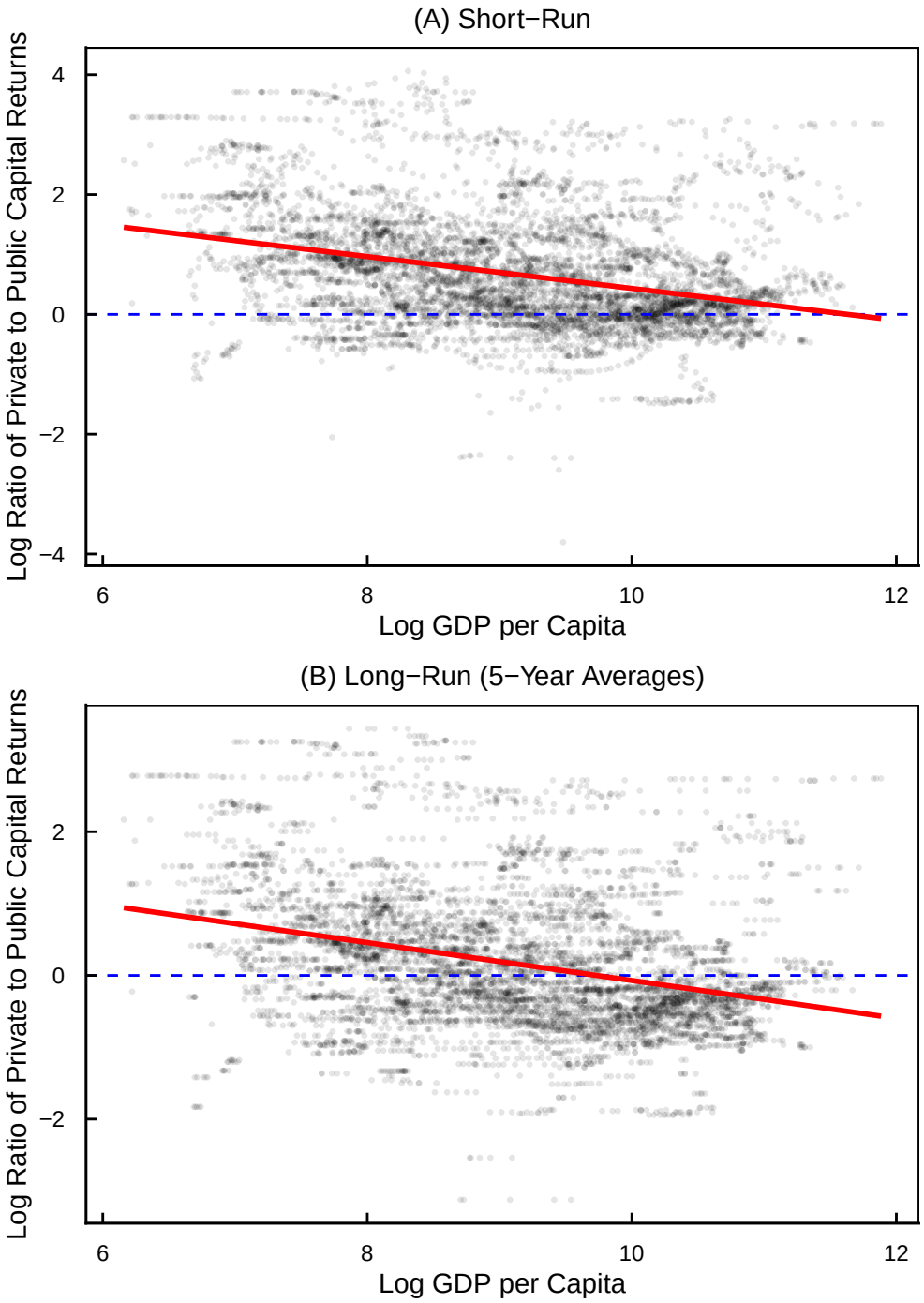
Recall Figure 2 that lower-income countries tend to have higher overall frictions, Figure 5 may suggest that the public capital investment frictions contribute less to the overall high frictions in

lower-income countries, and the private-side frictions are more likely to be the driving force.

3.4 Consumption Growth Marginal Sensitivity to Private vs. Public Frictions

4 Implications

Figure 5: Ratio of Private to Public Capital Returns by Log GDP per Capita



Notes: Panel (A) uses contemporaneous values with short−run output elasticity of public capital calibrated at 0.083. Panel (B) uses 5−year averages with long−run output elasticity of public capital calibrated at 0.122. Dashed line indicates ratio of 1 (log = 0). Red line is OLS fit.
Sources: PWT 11.0, ICSD.

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Appendix

Table A1: Polynomial Fit of Actual on Predicted Consumption Growth, with Controls

	Log Actual Consumption Growth		
	Linear	Quadratic	Cubic
Log Predicted Growth	0.082***		
	(0.015)		
Poly term 1		0.230***	0.230***
		(0.042)	(0.042)
Poly term 2		−0.250***	−0.250***
		(0.042)	(0.042)
Poly term 3			−0.033
			(0.042)
Log GDP Growth	0.110***	0.108***	0.108***
	(0.007)	(0.007)	(0.007)
Log TFP Growth	0.729***	0.730***	0.730***
	(0.009)	(0.009)	(0.009)
Intercept	0.014***	0.016***	0.016***
	(0.001)	(0.001)	(0.001)
Sample Size	5663	5663	5663
R^2	0.6463	0.6486	0.6486

* $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$

Notes: Orthogonal polynomial regressions of log actual on log predicted consumption growth, controlling for contemporaneous log GDP growth and log TFP growth. Standard errors in parentheses.

Table A2: Cross-Channel Granger Tests Across Specifications

	$\Delta \log f'(K^{priv})$ Short	$\Delta \log f'(K^{pub})$ Short	$\Delta \log f'(K^{priv})$ Long	$\Delta \log f'(K^{pub})$ Long	$\Delta \log K^{priv}$ Short	$\Delta \log K^{pub}$ Short	$\Delta \log K^{priv}$ Long	$\Delta \log K^{pub}$ Long
$\Delta \log f'(K^{pub}_{i,t-1})$	0.065**	0.145***						
	(0.024)	(0.017)						
$\Delta \log f'(K^{priv}_{i,t-1})$	−0.033	−0.040***						
	(0.018)	(0.012)						
$\Delta \log f'(K^{pub}_{i,w-1})$			0.001	0.306***				
			(0.051)	(0.045)				
$\Delta \log f'(K^{priv}_{i,w-1})$			0.168***	−0.037				
			(0.046)	(0.041)				
$\Delta \log K^{pub}_{i,t-1}$					0.087***	0.779***		
					(0.009)	(0.009)		
$\Delta \log K^{priv}_{i,t-1}$					0.749***	0.081***		
					(0.009)	(0.009)		
$\Delta \log K^{pub}_{i,w-1}$							0.015	0.451***
							(0.031)	(0.032)
$\Delta \log K^{priv}_{i,w-1}$							0.509***	0.177***
							(0.033)	(0.034)
$\Delta \log \text{TFP}_{it}$	0.605***	0.580***						
	(0.022)	(0.015)						
$\Delta \log \text{TFP}_{iw}$			0.864***	0.888***				
			(0.059)	(0.053)				
Sample Size	5704	5704	992	992	5704	5704	1079	1079
Within R^2	0.126	0.233	0.234	0.326	0.615	0.659	0.246	0.288

* $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$

Standard errors clustered at country level in parentheses. All specifications include country and year (window for long-run) fixed effects. Returns specifications include contemporaneous TFP control.

Table A3: Ratio

	Dependent Variable: $log(phi^{pub}/phi^{priv})$			
	Pooled	Country FE	Year FE	Country + Year FE
Log GDP per Capita	-0.266***	-0.060***	-0.299***	-0.138***
	(0.010)	(0.008)	(0.010)	(0.012)
Sample Size	6677	6677	6677	6677
Within R^2		0.009	0.120	0.021

* p $<$ 0.05, ** p $<$ 0.01, *** p $<$ 0.001

Standard errors clustered at country level in parentheses. $^{*}p < 0.05$, $^{**}p < 0.01$, $^{***}p < 0.001$.